Final Project Proposal: Multi-Agent System for Intelligent Tender Procurement

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Project Overview:

We propose to develop a multi-agent AI system that automates the discovery, analysis, and recommendation of government and private sector tenders for businesses. Our system will employ multiple specialized AI agents working collaboratively to search for relevant tenders, extract structured information, perform comparative analysis, and provide intelligent recommendations through an interactive dashboard and conversational interface.

Technical Approach:

Our system implements a multi-agent architecture with four specialized agents coordinating through a central orchestrator:

1. **Search & Download Agent:** Monitors tender portals (SAM.gov, state procurement systems) using web scraping and API integration. Given user-specified company profile (industry sector, capabilities, geographic scope, project size), this agent queries multiple sources and identifies relevant opportunities using keyword matching and relevance filtering.
2. **Document Processing Agent:** Extracts structured information using NLP techniques. We will use open-source transformer models from Hugging Face (LayoutLM or BERT-based models fine-tuned for information extraction). Key extractions include: project details, budget, deadlines, location, eligibility requirements, scope of work, and evaluation criteria using Named Entity Recognition and relation extraction.
3. **Analysis and Recommendation Agent:** Performs comparative analysis and matches tenders against company capabilities. This agent computes relevance scores based on geographic match, budget alignment, technical requirement overlap, competition level, and deadline feasibility using a weighted scoring system and collaborative filtering techniques.
4. **Conversational Interface Agent:** Provides natural language interaction for queries, clarifications, tender comparisons, and strategic advice using a retrieval-augmented generation (RAG) approach combining retrieved tender information with language model capabilities.

Agents communicate through a message-passing system, with the orchestrator managing task delegation, information sharing, and workflow coordination following FIPA standards or a custom protocol.

Implementation and Data

Implementation in Python using: Hugging Face Transformers for NLP models, Beautiful Soup and Scrapy for web scraping, PyPDF2 and python-docx for document parsing, scikit-learn for similarity computations, and open-source LLMs (Llama, Mistral) for conversational capabilities. Dashboard built using Streamlit or Flask with interactive visualizations.

We will collect 100-200 real tender documents from public procurement websites for training and validation. We will manually annotate 20-30 documents with key information fields to fine-tune our extraction models, then test on newly discovered tenders to demonstrate real-world applicability.

Evaluation and Metrics

Agent Performance: Precision and recall for Search Agent (relevant tenders found vs. irrelevant retrieved), information extraction accuracy for Document Processing Agent (F1 score for each field against manual annotations), and recommendation quality for Analysis Agent (user satisfaction, relevance of top-K recommendations).

System Integration: End-to-end processing time, inter-agent communication efficiency, and edge case handling (malformed documents, missing information).

Comparative Analysis: Benchmark against keyword-only search, manual document review, and single-agent processing to demonstrate multi-agent architecture advantages.